

Deep learning

1. *Develop the Neural Network*
2. *Able to calculate number of parameters*
3. *Write the math equations*
4. *Write the math equations in Matrix format*
5. *Definitions of*
 - *Forward prpogation*
 - *Back prpogation*
 - *Epoch*
6. *Forward prpogation is used to calculate the linear combination of weights and inputs
apply actiavation on top of that*
6. *Back prpogation is used to calculate the suitable parameters to reduce the error*
7. *Also we learned types of processing*
 - *Bacth process*
 - *Mini batch*
 - *Stochastic*

ask questions yourself

1. *What is actiavtion function formula*
2. *What are the equation*
3. *What is the ranges*
4. *What is the graph look like*
5. *What is the draw back*
6. *Where we will use?*

Activation functions

1. as we know every neuron has two opertions

- **Linear combination weights and inputs**
- **Activation**

2. Activation means we are introducing a Non linearity

3. If a neuron with out activations is a kind of a simple Linear equation

4. For more complex problems , a simple linear equation never works

5. In real time billions of neurons interconnected in many ways, every neuron transfer

a knowledge to another neuron

6. If that neuron not activated means, is simply using for transfer the information not analysing the information

7. Always think like this

**A complex problem to solve, is a human brain feeling =
Non linear equations in maths**

we have many activation functions are there

1. Sigmoid

2. Softmax

3. Tanh

4. ReLU (Rectified linear unit)

5. Leaky ReLU

6. GELU (Gaussian Error Linear Unit)

The questions need to be address

1. What is the activation function? what it is useful

2. Name some important activation function

3. What is formulae of activation function

4. What is the range of activation function and Graph

5. what is the differentiation of activation function

*6. What is the **Vansih gradients problem***

7. Which activation function will use in Hidden layer

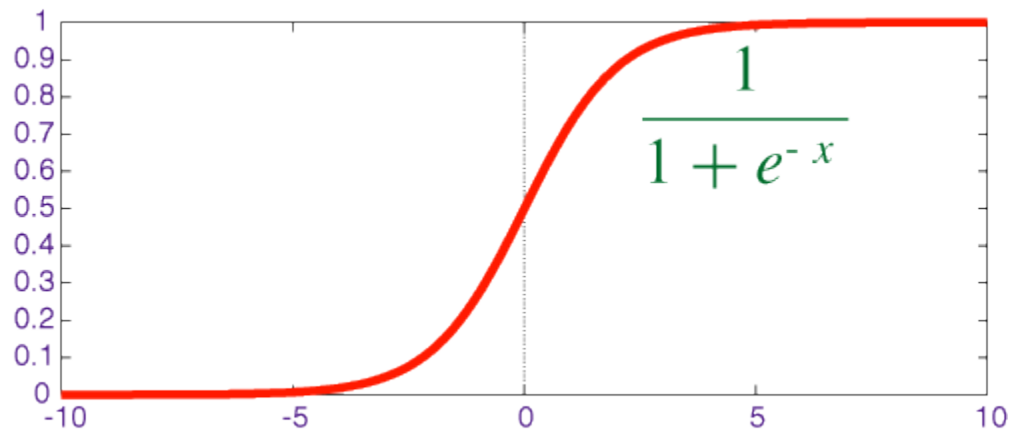
8. Which activation function will use in Output layer

Sigmoid

$$\text{Formulae: } s = \frac{1}{1 + e^{-x}}$$

Range = 0 to 1

Graph:



Differentiation:

$$\frac{u}{v} = \frac{u'v - v'u}{v^2}$$

$$\frac{u}{v} = \frac{1}{1 + e^{-x}}$$

$$u = 1, v = 1 + e^{-x}$$

$$u' = 0, v' = -1 * e^{-1*x}$$

$$\frac{u'v - v'u}{v^2} = \frac{0 * (1 + e^{-x}) - (-1 * e^{-x}) * 1}{(1 + e^{-x})^2} = \frac{e^{-x}}{(1 + e^{-x})^2}$$

$$s = \frac{1}{1 + e^{-x}}$$

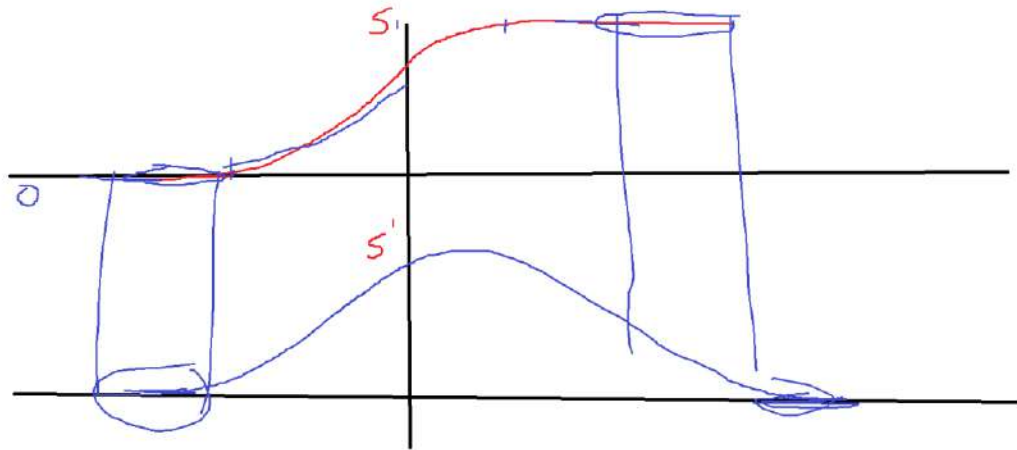
$$s' = \frac{1 + e^{-x} - 1}{(1 + e^{-x})^2} = \frac{1 + e^{-x}}{(1 + e^{-x})^2} - \frac{1}{(1 + e^{-x})^2}$$

$$\frac{1 + e^{-x}}{(1 + e^{-x})^2} - \frac{1}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \left[\frac{1 + e^{-x}}{1 + e^{-x}} - \frac{1}{1 + e^{-x}} \right]$$

$$\frac{1}{1 + e^{-x}} \left[1 - \frac{1}{1 + e^{-x}} \right]$$

$$s[1 - s] = s(1 - s)$$

So the derivation of sigmoid function is : $s(1 - s)$



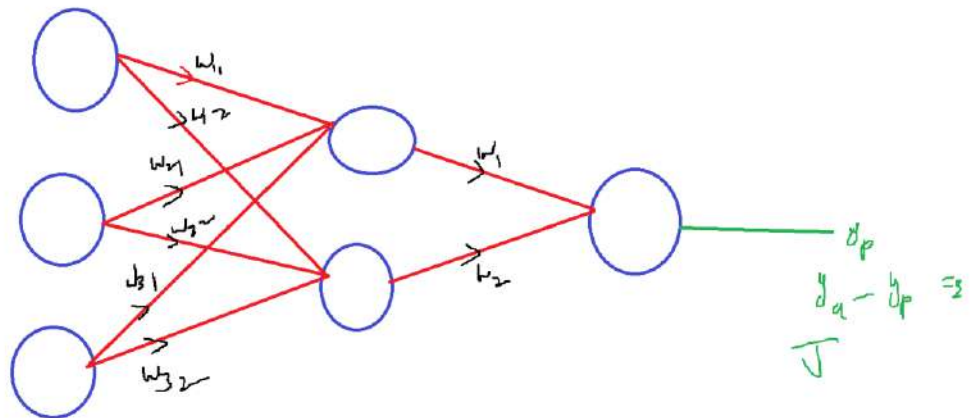
*s' curve at the edges leads to zero , because in sigmoid the value ranges 0 to 1
differentiation of zero and one is euqal to zero*

Where we are using the differentiation in NN:

we are using the differentiation in Grdaient Descent algorithm

GD is used to find the suitable weights

$$w_{new} = w_{old} - \alpha * \frac{\partial J}{\partial w} |_{w=w_{old}}$$



$$w_{1_{new}} = w_{1_{old}} - \alpha * \frac{\partial J}{\partial w_1} |_{w_1 = w_{1_{old}}}$$

$$w_{11_{new}} = w_{11_{old}} - \alpha * \frac{\partial J}{\partial w_{11}} |_{w_{11} = w_{11_{old}}}$$

$$\frac{\partial J}{\partial w_{11}}$$

$$\text{output } (J) = \text{activation}(w_0 + h_1 * w_1 + h_2 * w_2)$$

$$h_1 = \text{activation}(x_1 * w_{11} + x_2 * w_{12} + x_3 * w_{13})$$

$$\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial h_1} * \frac{\partial h_1}{\partial w_{11}}$$

$$y = z^2 + 1$$

$$z = 2x + 2$$

$$\frac{dy}{dx} = \frac{dy}{dz} * \frac{dz}{dx}$$

$$\frac{dy}{dz} = 2z; \frac{dz}{dx} = 2$$

$$\frac{dy}{dx} = 2z * 2 = 4z$$

$$y = z^2 + 1$$

$$z = 2x + 2$$

$$x = 2k - 2$$

$$\frac{dy}{dk} = \frac{dy}{dz} * \frac{dz}{dx} * \frac{dx}{dk}$$

$$\frac{\partial J}{\partial h_1} = \text{we are differentiating activation function}$$

when we are differentiation sigmoid for 0 and 1, that derivation = 0

$$\frac{\partial J}{\partial w_n} = \frac{\partial J}{\partial h_n} * \frac{\partial h_n}{\partial w_{n-1}} * \frac{\partial w_{n-1}}{\partial h_{n-1}} * \dots * \frac{\partial h_o}{\partial w_n}$$

$$\text{assume that some where } \frac{\partial w_{n-1}}{\partial h_{n-1}} = 0$$

$$\frac{\partial J}{\partial w_n} = 0$$

$$w_{1_{new}} = w_{1_{old}} - \alpha * 0$$

$$w_{1_{new}} = w_{1_{old}}$$

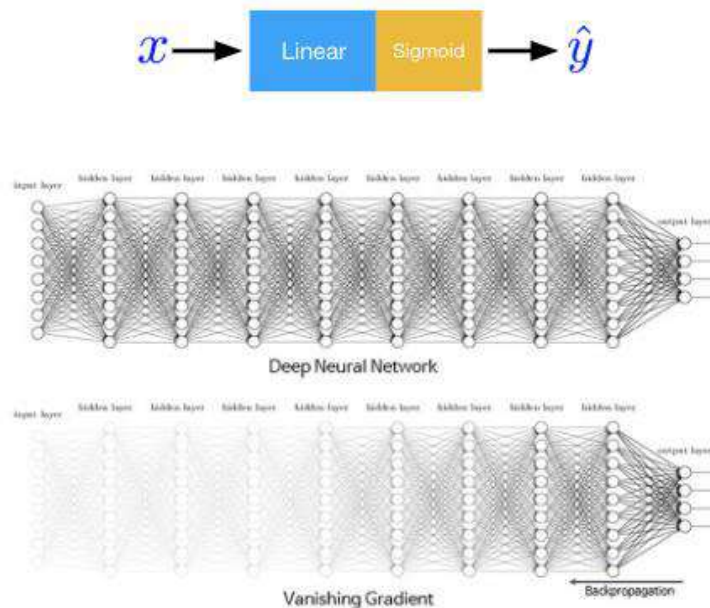
the weights are not updating , new weights = old weights

no learning

model become dummy model

This problem is called as Vanishing gradients problem

Sigmoid: Vanishing Gradient Problem



Vanishing gradients

- *Main goal is to find the suitable coeff*
- *we are finding by using of Gradient Descent algorithm*
- *In GD we are differentiation of cost function $J = (y_{actula} - y_{pred})^2$*
- *$y_{prediction}$ means output of NN (output of Neurons)*
- *Each neuron is activation of linear combination of input*
- *assume that we are using sigmoid activation function in hidden layer*
- *we are differentiating of activation function*
- *Differentitaon of sigmoid activation function become zero for low and high values*
- *this leads to J become zero*
- *$W_{new} = W_{old}$*
- *new weighths = old weights*
- *weights are not updating*
- *model will not learn anything, the leraning will stop*
- *this is called as Vanishing Gradients problem*
- *Thats why sigmoid activation function will not use in hidden layers*
- *Sigmoid activation function will use at output layer that to bi classification*

1. Formula

2. Range of the values

3. Graphical view

4. Differentiation

5. Vanish Gradient problem occurs or not

6. in which layer we will use

sigmoid vs Softmax vs tanh

. ReLU vs Leaky ReLU